

Justin's Portfolio

Justin Burton

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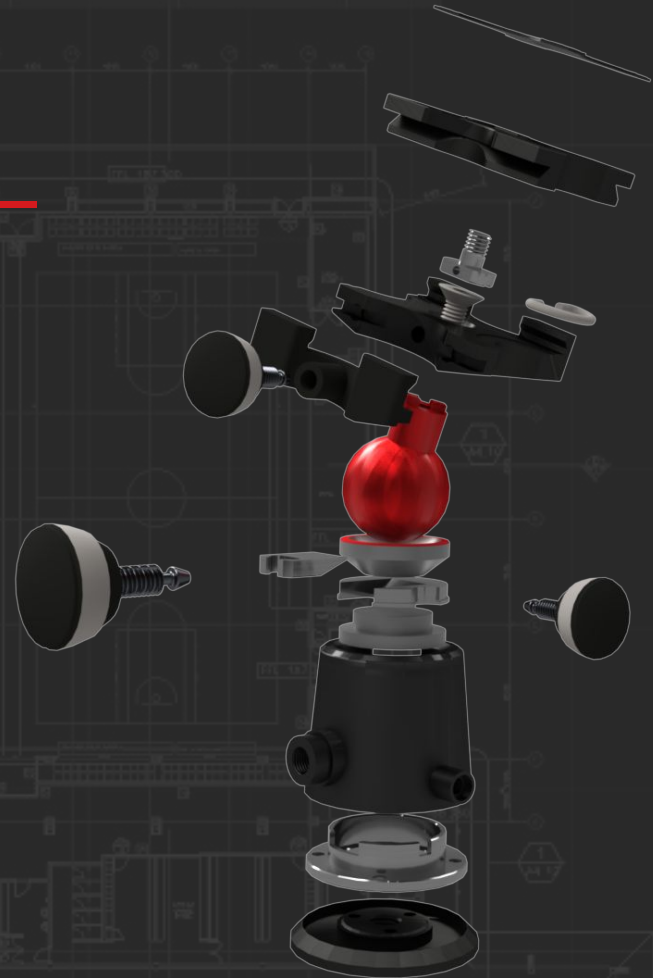
Joby Ball Head Photography Attachment

Rubber Grip

Ball Joint

Body

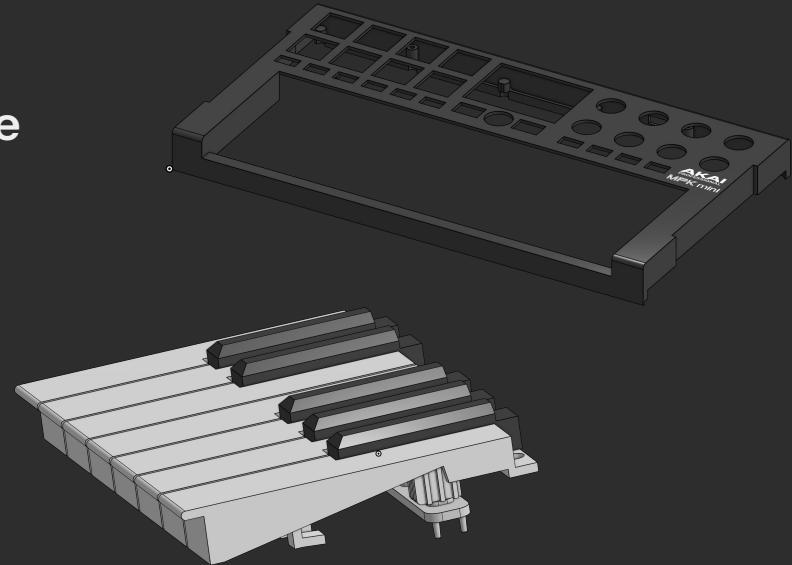
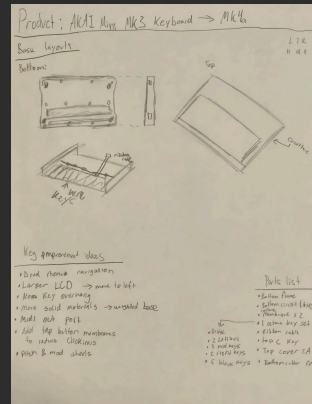
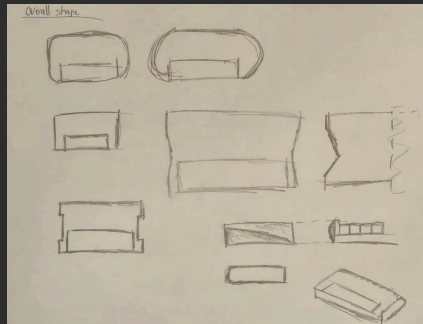
Labeled parts modeled
based on provided
drawings. Assembly
created using additional
provided parts



AKAI MPK Mini MK3 Next Generation Product

Task: Evaluate feedback, ratings, and reviews of the AKAI MPK Mini MK3 and develop a next generation product implementing improvements.

Process: Sketch concepts, develop parts in CAD, import electronic components, create assembly



CAD Analysis

Interference detection ✕

Instances to analyze

473340001 <5>	✕
74LVC245 <22>	✕
74LVC245 <2>	✕
74LVC245 <8>	✕
74LVC245 <12>	✕
74LVC245 <14>	✕
74LVC245 <10>	✕
74LVC245 <18>	✕
74LVC245 <3>	✕
74LVC245 <19>	✕
74LVC245 <15>	✕
74LVC245 <16>	✕
74LVC245 <21>	✕

☐ Include standard content

☐ Show top level only

Interferences

EVW-AE4001B14 <6> and ...

UpperPCB <1> and 1770 <4>

UpperPCB <1> and SS-524...

↺

?

Part Face

Instances to measure

473340001 <5>	✕
74LVC245 <22>	✕
74LVC245 <3>	✕
74LVC245 <18>	✕

Mate connector for reference frame

☐ Show calculation variance

Mass ☐ Override

1.815 lb

Volume

41.107 in³

Surface area

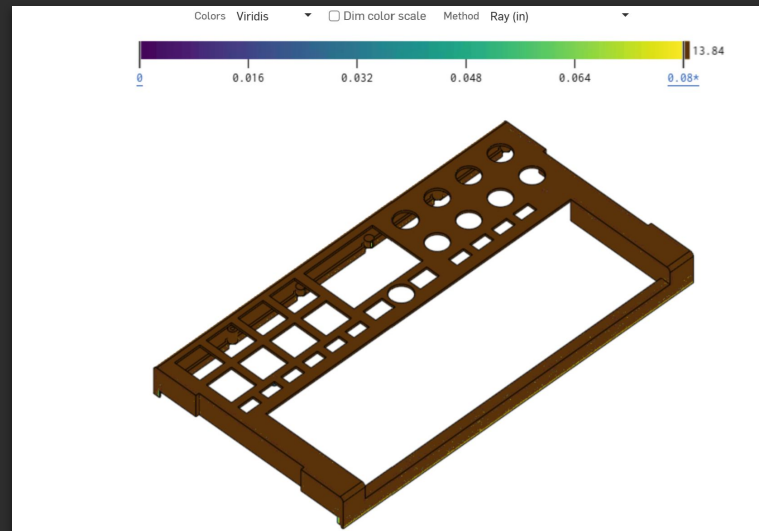
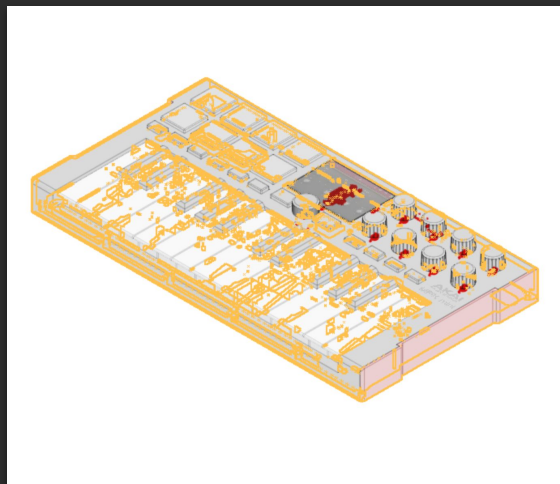
1048.024 in²

Center of mass ☐ Override

X ↘ 6.941 in

Y ↗ 4.299 in

Z ↑ 0.729 in



Measure ✕

Select entities to measure

Face of Potentiometer <1>	✕
Face of Base <1>	✕

Measure type: Show all

Length unit: Inch

Angle unit: Degree

☐ Reference coordinate system

Parallel dist: 1.680 in

X ↘ 0.000 in

Y ↗ 0.000 in

Z ↑ 1.680 in

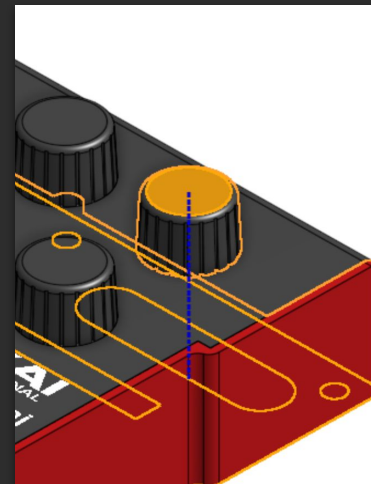
Area: 81.013 in²

Min dist: 1.680 in

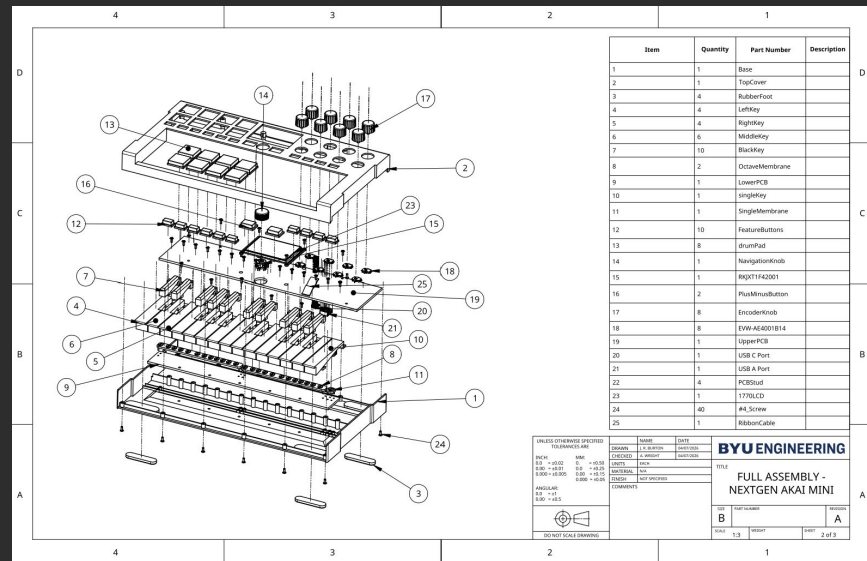
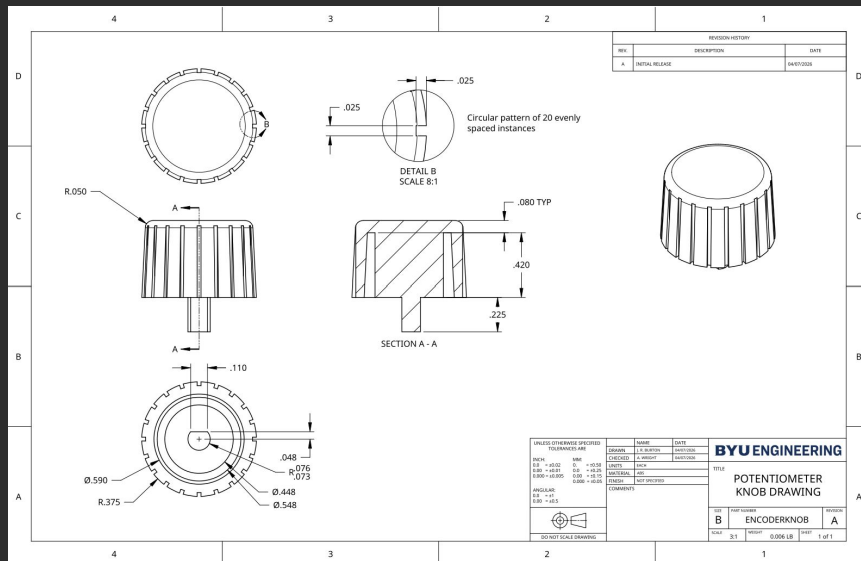
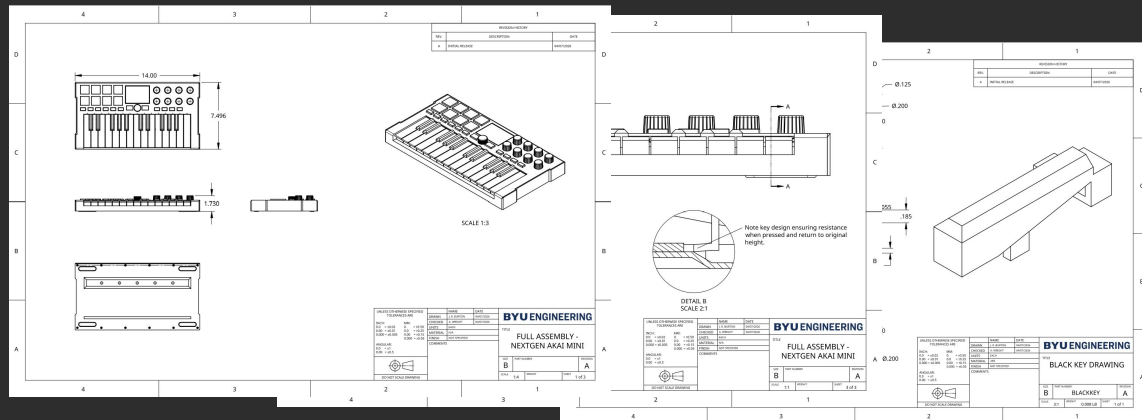
X ≠ 0.000 in

Y ≠ 0.000 in

Z ≠ 1.680 in



Engineering Drawings

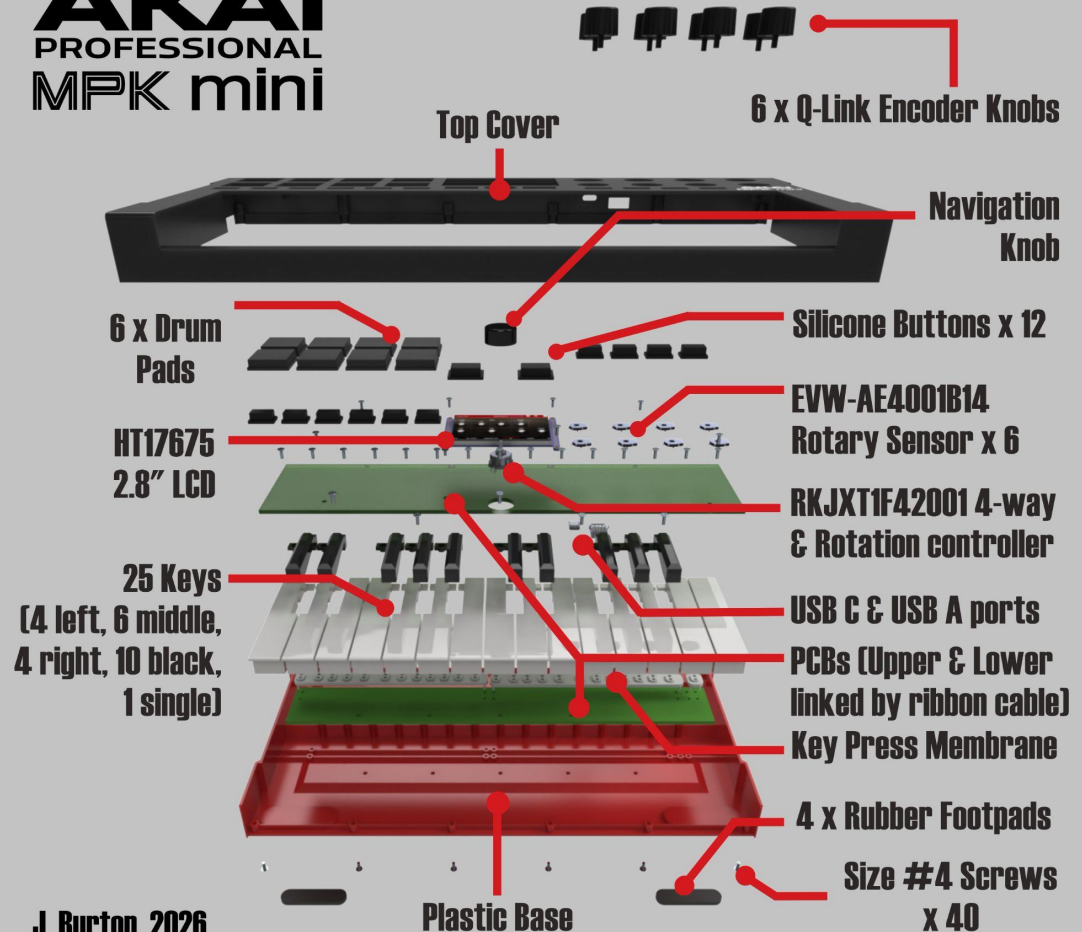


Professional Renders



USB ports, rotary sensors, LCD and
4-way controller imported

AKAI
PROFESSIONAL
MPK mini



J. Burton, 2026

OLD



Updated Intuitive layout

NEW



Larger LCD display

Improved navigation control

Sturdier Build

Wisma 46 Lego Model

Configurations

Brick Color **White**
Block Type **Plate**
Length 4
Width 2
scale 0.397
Tube Type **Tubes**
☐ Decal

Filter by name or type

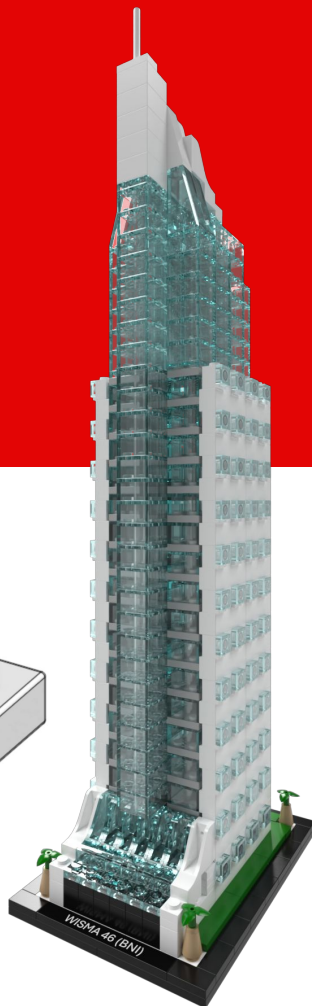
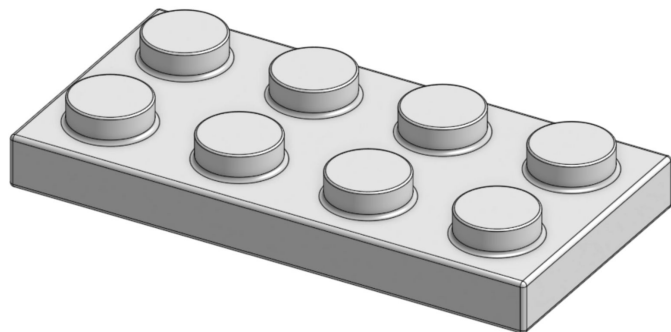
Features (23)

Default geometry

- Origin
- Top
- Front
- Right

(x) #Height = 1

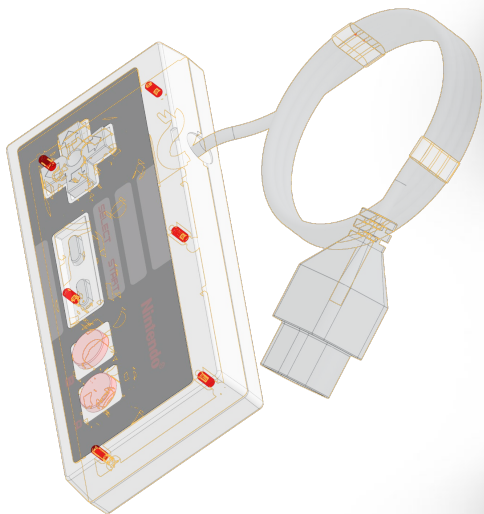
(x) #Unit = 20 mm



Built with custom configurable
brick generator



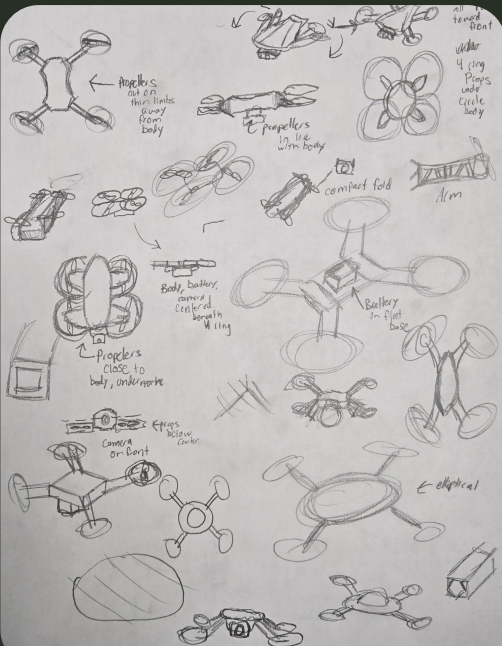
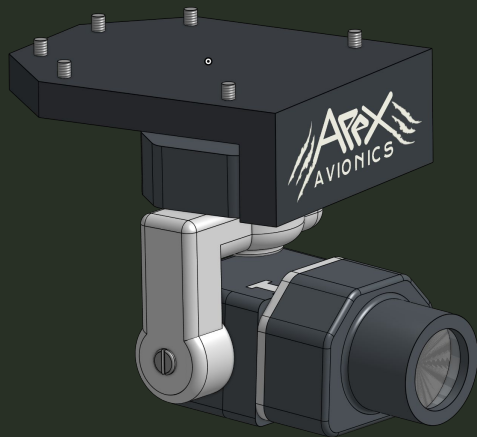
NES Controller & Configurable Grip Case



Cable and some internal parts
imported for assembly with parts
modeled by me



Folding Photography Drone



THE ALL NEW RAPTOR ONE



Swappable Cameras

Use lens you want
for the shot you need

Lightweight & Compact

Drone arms fold in
for easy travel

3 Axis Gimble System

Ensures stable videos
and crystal clear photos

Arms/Motor modeled by Kael Wagner
Body partially modeled by Ben Linford

RAPTOR ONE PITCH

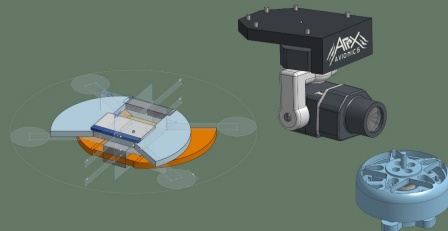
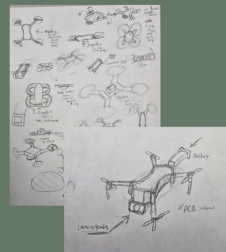
Feasibility:

This design is centered on feasibility and ease of production. This was ensured with the following standards:

- The design is mainly shells allowing for production by injection molding.
- Product utilizes standard screws and existing electronic parts.
- The design is lightweight and uses propellers of the proper size allowing for flight
- The product is designed to be easily assembled by hand



Design Progress:



Initial Sketches → Top Down Definitions → CAD Development

Technical Requirements:

Design: Unique quadrotor design with arms that fold or retract into the body
Propulsion: 4-inch radius propellers and 1-inch diameter brushless motors.

Power & Storage: Removable LiPO battery (0.380" x 1.010" x 2.220"), 12 in³ main PCB, and accessible microSD slot.

Gimbal Camera: 3-axis motion (X, Y, Z) with transparent lens/cover, body-protected yet unobstructed.

Manufacturing: Injection-molded body/arms (0.020"-0.060" wall thickness) using top/bottom shells joined by at least 4 screws.

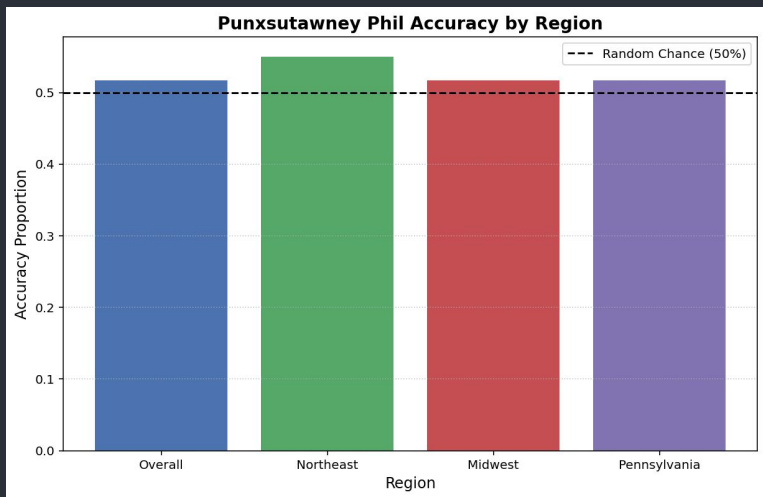
Weight: Total assembly mass must be 53% of competitive drone weights.

This design meets all of the above technical requirements



Statistical Analysis

Goal: Evaluate data from the Punxsutawney Groundhog Club and NOAA to determine if the Groundhog Day groundhog's "weather forecasts" are more or less accurate than random chance



```
import pandas as pd
import numpy as np
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt

df = pd.read_csv('groundhogData.csv')

df = df[df['Punxsutawney Phil'] != 'No Record'].copy()

df['Feb_Mar_Avg'] = (df['February Average Temperature (Pennsylvania)'] +
                    df['March Average Temperature (Pennsylvania)']) / 2

threshold = df['Feb_Mar_Avg'].median()

df['Actual_Spring'] = np.where(df['Feb_Mar_Avg'] > threshold, 'Early Spring', 'Late Spring')

def check_accuracy(row):
    prediction = row['Punxsutawney Phil']
    actual = row['Actual_Spring']
    if (prediction == 'No Shadow' and actual == 'Early Spring') or \
        (prediction == 'Full Shadow' and actual == 'Late Spring'):
        return 1 # Correct
    return 0 # Incorrect

df['Is_Correct'] = df.apply(check_accuracy, axis=1)

# one tail test
successes = df['Is_Correct'].sum()
nobs = len(df)
null_prop = 0.5

#z stat & p value
z_stat, p_value = proportions_ztest(count=successes, nobs=nobs, value=null_prop, alternative='larger')

#95% conf
from statsmodels.stats.proportion import proportion_confint
ci_low, ci_high = proportion_confint(successes, nobs, alpha=0.05, method='normal')

print(f"Total Samples (n): {nobs}")
print(f"Correct Predictions: {successes}")
print(f"Sample Proportion: {successes/nobs:.4f}")
print(f"P-value: {p_value:.4f}")
print(f"95% Confidence Interval: ({ci_low:.4f}, {ci_high:.4f})")
```

Evaluating Predictive models of Human Behavior in Engineering Design

Academic Research conducted as part of the BYU Design Exploration Research Group.

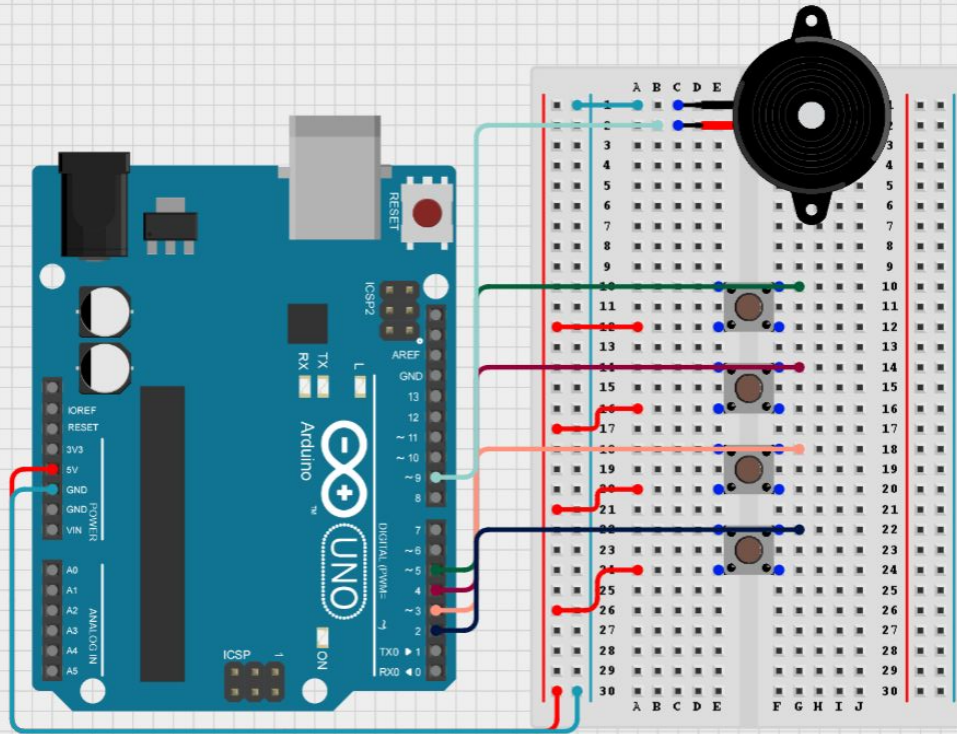
Explored Predictive models of human behavior and their applications to the engineering design process.

The findings have been presented at 2 academic conferences and published in the conference proceedings.

An additional paper on the topic will be submitted to journal this month.

Stage / Sub-Task (Required Information)	PA	ABM	EBM	DES	GBM	CAM	HBM
Stage 1: Planning and Task Clarification	11	8	8	9	9	7	11
Gather stakeholder needs & data patterns [91]	4	2	3	3	3	1	4
Define measurable requirements & goals [42]	4	2	4	3	2	2	3
Identify human constraints & contexts [91]	3	4	1	3	4	4	4
Stage 2: Conceptual Design	7	12	8	9	11	9	9
Generate alternative conceptual solutions [91]	3	2	2	2	2	2	3
Model interactions & social dynamics [91]	2	5	3	3	5	2	3
Evaluate behavior related to candidate concepts [42]	2	5	3	4	4	5	3
Stage 3: Embodiment Design	5	9	10	7	8	10	7
Translate concept into preliminary system layout [30]	1	1	1	1	1	1	1
Model operational dynamics & interactions [30]	2	4	4	4	4	4	3
Analyze concept behavior & feedback loops [30]	2	4	5	2	3	5	3
Stage 4: Detail Design & Verification	7	11	10	11	9	13	7
Refine detailed interactions/decision outcomes [91]	2	4	3	3	3	4	2
Validate system behavior against criteria [42]	3	4	4	4	3	4	3
Finalize human-behavior model integration [91]	2	3	3	4	3	5	2
Stage 5: Design Analysis & Testing	7	12	9	11	10	10	6
Performance verification through simulation [65]	2	5	3	3	3	4	2
Reliability analysis & failure mode identification [105]	2	2	3	5	2	4	2
Lifecycle analysis & impact prediction [2]	3	5	3	3	5	2	2

One Key Keyboard



Parts List:

Arduino Uno

4 x Push button

Passive Buzzer

14 connector wires

Autonomous Task-Specific Robot

